

Isabel Moore

US Citizen · [REDACTED] · College Station, TX

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EDUCATION

Texas A&M University

Master of Science in Computer Engineering

Graduation: May 2026

Thesis: Meta Reinforcement-Learning based Contextual Adaptive-Control

Bachelor of Science in Mechanical Engineering

Graduation: May 2024

EXPERIENCE

Systems Design Intern [REDACTED] Team

May 2025 - Present

Northrop Grumman [REDACTED], [REDACTED]

- Optimized initial conditions for [REDACTED], supporting both 6DOF and 3DOF models within [REDACTED] framework.
- Trained LSTM and Transformer models for [REDACTED] prediction, achieving [REDACTED] error through parallel execution and SLURM-based deployment for simulation generation.

Controls & Resilient Networks Researcher

May 2024 - Present

Bush Combat Development Complex

Bryan, TX

- Designed full meta-RL training pipeline as core thesis research, enabling contextual adaptive-control that generalizes across robot platforms and surface conditions, successfully bridging sim-to-real deployment.
- Achieved < 5cm cross-track error across 50m path tests by optimizing policy robustness over friction coefficients ranging from 0.01μ to 1.0μ .

Computer Vision Researcher

Aug 2024 - May 2025

Computer Vision & Robotics Laboratory

College Station, TX

- Developed open-source data curation pipeline (Mcity Data Engine) integrating vision foundation models (Depth Pro, SAM2) to auto-identify high-value training samples, reducing manual labeling for perception research.

Guidance, Navigation, and Controls (GNC) Researcher

Sep 2023 - May 2024

Connected Autonomous Safe Technologies (CAST) Laboratory

College Station, TX

- Developed Kalman Filter (KF) state estimation framework to mitigate GPS spoofing attacks, reducing localization errors to less than 3% under spoofed conditions.
- Engineered Dead-Reckoning navigation system for off-road AVs, reducing yaw error from 180° to $2-8^\circ$ (97% reduction).

Team Lead

Aug 2023 - Jun 2024

General Motors-SAE AutoDrive Challenge™ II

Ann Arbor, MI

- Led 6-person team in path planning and ADAS optimization, securing 2nd place in international competition.
- Enhanced Pure-Pursuit lateral control algorithms, enhancing vehicle stability and reducing path error by 70%.

PUBLICATIONS & OPEN SOURCE WORK

- Mcity Data Engine: Iterative Model Improvement Through Open-Vocabulary Data Selection, IEEE Intelligent Transportation Systems (ITSC), Gold Coast, Australia, 2025. [paper] [GitHub]
- Using Sensor-Health-Aware Resilient Fusion for Localization in the Presence of GPS Spoofing Attacks, IEEE International Conference on Cyber Security and Resilience (CSR), London, United Kingdom, 2024. [paper]

PROJECTS

- Meta-RL 6DOF [REDACTED] Guidance: Gymnasium environment for [REDACTED] simulation; trained RecurrentPPO with LSTM for LOS-rate-based guidance.
- SatTrack: Real-time satellite tracking app with orbital propagation and health monitoring. [Link]
- Meta-RL Differential Drive Navigation: Trained PPO-based agent for robot-agnostic navigation robust to varying friction conditions, validated across Clearpath Jackal and Warthog platforms.

TECHNICAL SKILLS

- Programming Languages:** Python, C/C++, Java, Shell, SQL/SQLite
- Developer Tools & Frameworks:** Git, Docker, ROS1/ROS2, Linux, SLURM, TensorFlow, PyTorch, Voxel51
- Simulation/Design:** Gazebo, RViz, CARLA, LabVIEW, MATLAB/Simulink, SolidWorks, ANSYS, AutoCAD